

Exploratory Data Analysis and Predictive Modeling with Synthetic Splicing Dataset

Department of Health Informatics
MS Health Informatics
BINF 5040 Biomedical Information Processing

Project Assignment

By

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Overview and Objective

Overview of Gene-Splicing Data: a synthetic dataset containing the expressions of three splicing factors and a related splicing event.

Objective: Perform exploratory data analysis (EDA), data visualization, and predictive modeling to gain insights into the relationships between splicing factors and the splicing event.

Outline

Overview and Objective

1. Data Exploration

1.1 Summary statistics of the dataset

1.2 Check for missing values and outliers.

1.3 Correlation matrix to assess relationships

2. Data Visualization

2.1 Histograms to show the distribution of splicing factors and splicing events

2.2 Scatterplot to visualize the relationships between each individual splicing factor and the splicing event

3. Predictive Modeling

3.1 Splitting the dataset into training and testing sets.

3.2 Building a predictive training model using linear regression.

3.3 Evaluation of model performance using appropriate metrics (e.g., R-squared, Mean Squared Error) on the testing set.

3.4 Interpret the model's coefficients to understand the relationships between the splicing factors and the splicing event.

4. Conclusions and Recommendations

4.1 Summary of key findings from EDA and modeling.

4.2 Identification of splicing factors most strongly associated with the splicing event.

4.3 Recommendations for further analysis or experiments based on results.

1. Data Exploration

1.1 Summary statistics of the dataset

splicing_data.csv

> str(splicingData)

'data.frame': 100 obs. of 5 variables:

```
$ SubjectID: int 1 2 3 4 5 6 7 8 9 10 ...
$ SplicingFactor1: num 4.44 4.77 6.56 5.07 5.13 ...
$ SplicingFactor2: num 6.93 8.39 7.63 7.48 6.57 ...
$ SplicingFactor3: num 5.76 5.05 3.79 4.43 3.67 ...
$ SplicingEvent : num -0.0735 -2.0188 1.6893 -0.9648 1.3597 ...
```

> summary(splicingData)

SubjectID	SplicingFactor1	SplicingFactor2	SplicingFactor3	SplicingEvent
Min. : 1.00	Min. :2.691	Min. : 4.920	Min. :2.595	Min. : -7.6823
1st Qu.: 25.75	1st Qu.:4.506	1st Qu.: 6.798	1st Qu.:3.575	1st Qu.: -2.0259
Median : 50.50	Median :5.062	Median : 7.661	Median :4.029	Median : 0.7102
Mean : 50.50	Mean :5.090	Mean : 7.839	Mean :4.096	Mean : 0.3985
3rd Qu.: 75.25	3rd Qu.:5.692	3rd Qu.: 8.702	3rd Qu.:4.611	3rd Qu.: 2.5093
Max. :100.00	Max. :7.187	Max. :12.862	Max. :5.834	Max. : 7.4030

	SubjectID	SplicingFactor1	SplicingFactor2	SplicingFactor3	SplicingEvent
1	1	4.439524	6.934390	5.759048	-0.07349676
2	2	4.769823	8.385326	5.049930	-2.01875607
3	3	6.558708	7.629962	3.787884	1.68933793
4	4	5.070508	7.478686	4.434555	-0.96476130
5	5	5.129288	6.572572	3.668528	1.35966220
6	6	6.715065	7.932458	3.619002	4.00330194
7	7	5.460916	6.822643	3.369118	-1.65599466
8	8	3.734939	5.498087	3.524306	1.40886085
9	9	4.313147	7.429660	5.320726	2.61551705
10	10	4.554338	9.378495	3.956777	1.09447048
11	11	6.224082	7.136980	4.095396	6.39274434
12	12	5.359814	8.911946	4.194950	0.96273243
13	13	5.400771	5.573176	4.985981	1.48130855
14	14	5.110683	7.916657	3.587149	-1.06305909
15	15	4.444159	8.779111	3.205994	-3.38144425
16	16	6.786913	8.451730	5.340558	4.97355779
17	17	5.497850	8.158514	3.647069	0.37012156
18	18	2.022202	7.028041	2.421547	5.20816044

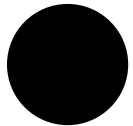
1.2 Check for missing values

```
> sum(is.na(splicingData))
```

OUTPUT:

```
[1] 0
```

RESULTS: No missing values were found



1. Data Exploration

1.3 Check for outliers

```
splicingData_Boxplot <-boxplot(splicingData[2:5])
```

```
print(splicingData_Boxplot)
```

```
OUTPUT:> splicingData_Boxplot <-boxplot(splicingData[2:5])
```

```
> print(splicingData_Boxplot)
```

```
$stats
```

	[,1]	[,2]	[,3]	[,4]
[1,]	3.033383	4.920129	2.594778	-7.6823337
[2,]	4.503323	6.774043	3.566803	-2.0290885
[3,]	5.061756	7.661255	4.028732	0.7101694
[4,]	5.694998	8.726275	4.632604	2.5273547
[5,]	7.187333	11.192678	5.834463	7.4030262

```
$n
```

```
[1] 100 100 100 100
```

```
$conf
```

	[,1]	[,2]	[,3]	[,4]
[1,]	4.873472	7.352802	3.860335	-0.009748581
[2,]	5.250041	7.969707	4.197128	1.430087460

```
$out
```

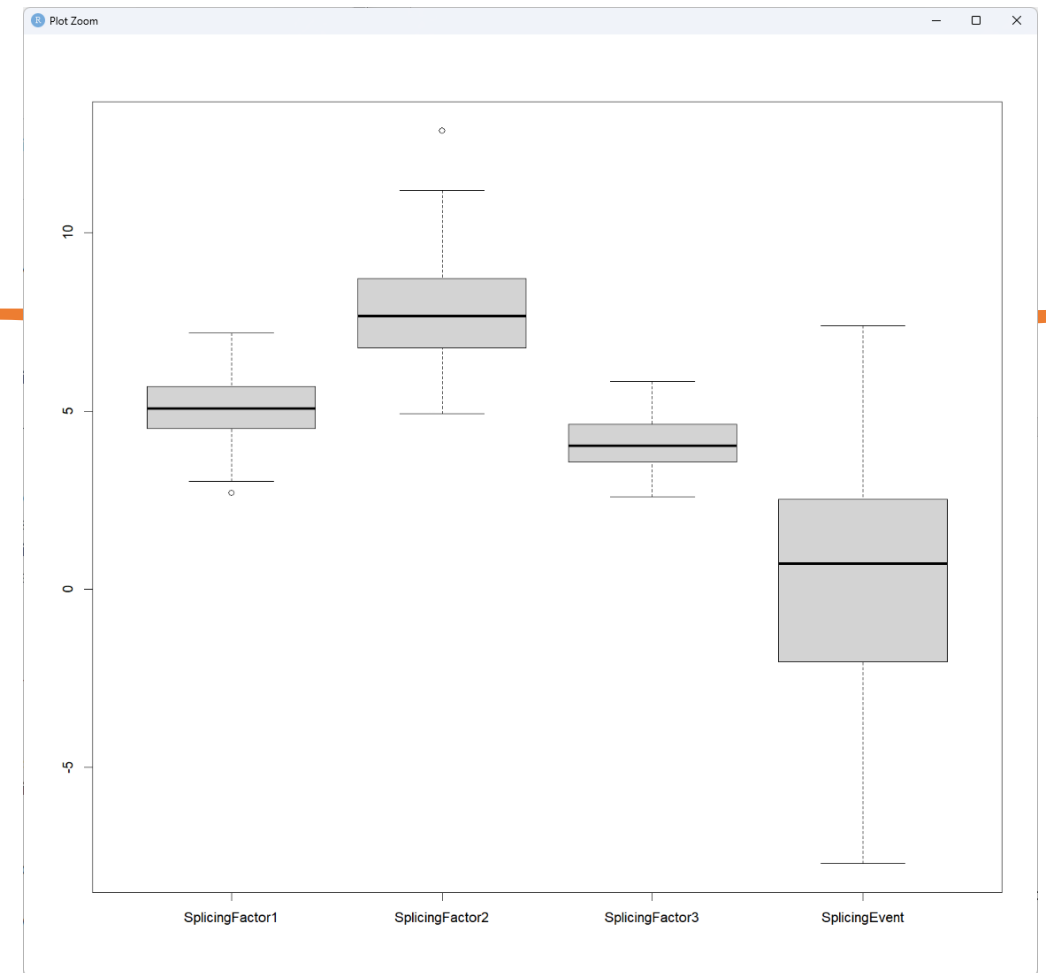
```
[1] 2.690831 12.861560
```

```
$group
```

```
[1] 1 2
```

```
$names
```

```
[1] "SplicingFactor1" "SplicingFactor2" "SplicingFactor3" "SplicingEvent"
```



RESULTS: Using the Box plot graph, two outliers were observed:

-Splicing Factor 1 (2.690831)

-Splicing Factor 2 (12.861560)

1. Data Exploration

1.4 Correlation matrix to assess relationships

Null Hypothesis: There is no relationship between Splicing Factors and the Splicing Event.

```
Using rcor()- pearson
library(Hmisc)
```

```
rcorr(splicingDataMatrix)
```

Output:

```
> rcorr(splicingDataMatrix)
```

	SplicingFactor1	SplicingFactor2	SplicingFactor3	SplicingEvent
SplicingFactor1	1.00	-0.05	-0.13	0.52
SplicingFactor2	-0.05	1.00	0.03	-0.63
SplicingFactor3	-0.13	0.03	1.00	0.00
SplicingEvent	0.52	-0.63	0.00	1.00

n= 100

p

	SplicingFactor1	SplicingFactor2	SplicingFactor3	SplicingEvent
SplicingFactor1		0.6246	0.2002	0.0000
SplicingFactor2	0.6246		0.7626	0.0000
SplicingFactor3	0.2002	0.7626		0.9643
SplicingEvent	0.0000	0.0000	0.9643	

Interpretation:

SplicingFactor1 is **positively** related to the Splicing event with a Pearson correlation coefficient of **r = 0.52**, which is a large effect. The significance value, p of **0**(< .001) indicates that there is **zero** probability that the null hypothesis is true.

SplicingFactor2 is **negatively** related to the splicing event, with a Pearson correlation coefficient of **r= -0.63**, which is a large effect. The significance value, p of **0**(< .001) indicates that there is **zero** probability that the null hypothesis is true.

SplicingFactor3 has **no relation** with the Splicing event with a Pearson correlation coefficient of **r= 0.00**. The significance value, p of **0.9643**(> .001) indicates that there is 96% probability that the null hypothesis is true.

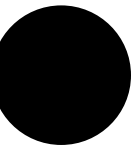




2. Data Visualization

2.1 Histograms with curve to show the distribution of splicing factors and splicing events

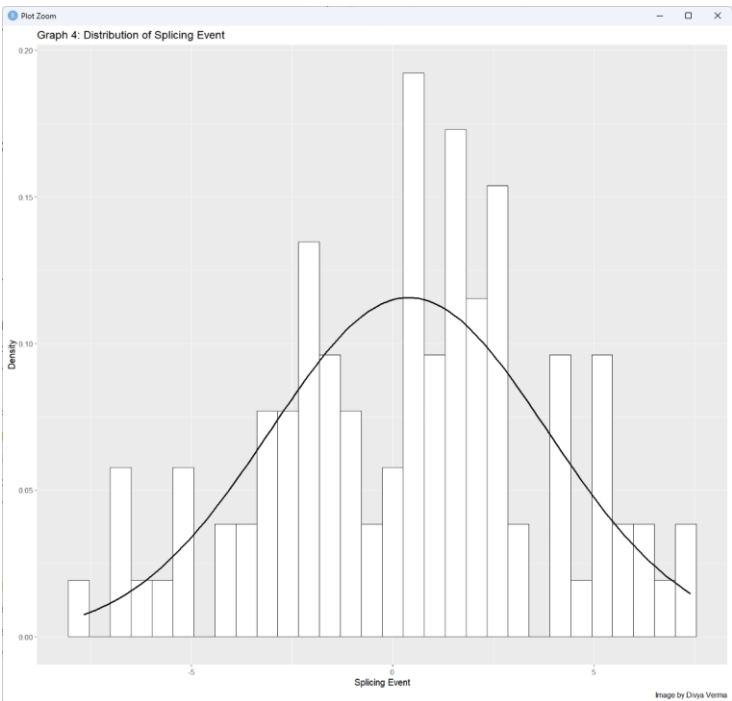
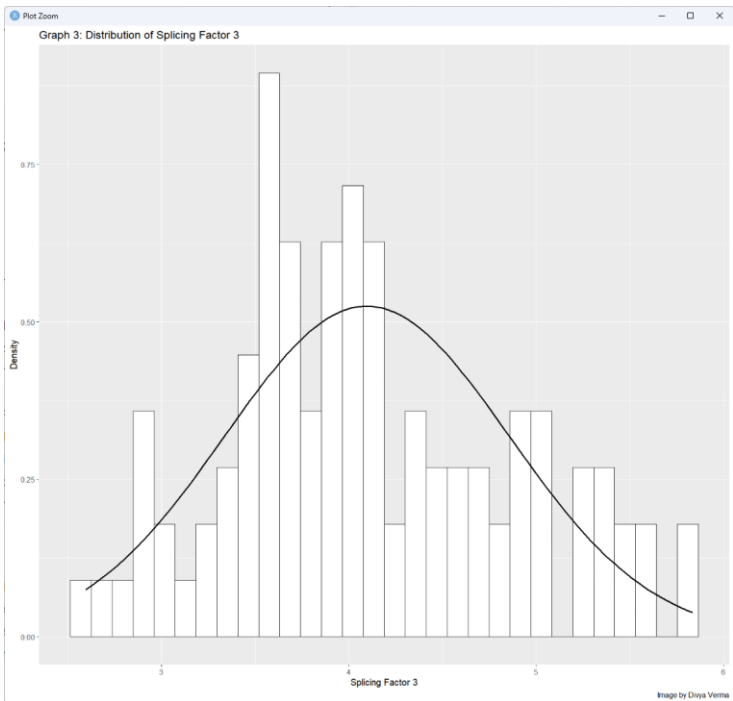
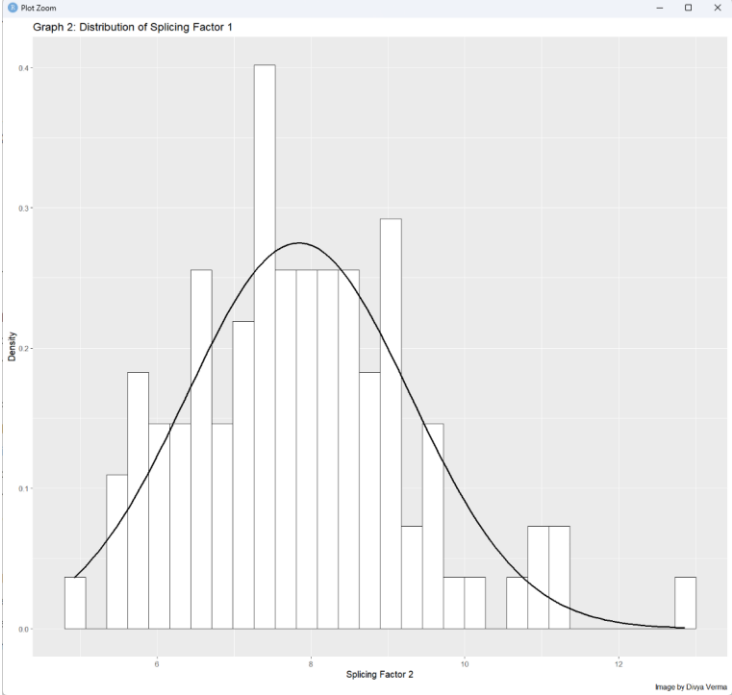
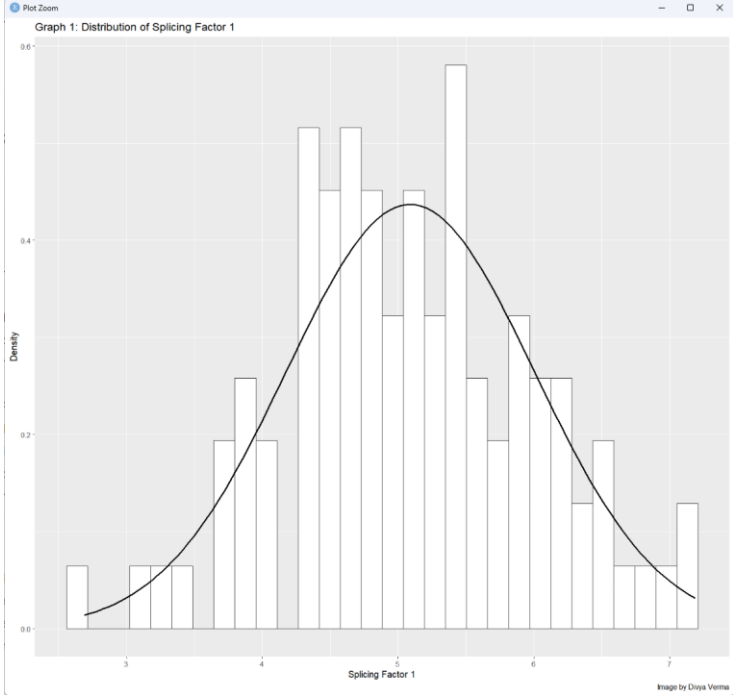
2.2 Scatterplot to visualize the relationships between each individual splicing factor and the splicing event



2. Data Visualization



2.1 Histograms with normal curve to show the distribution of splicing factors and splicing events



2. Data Visualization

Quantifying Normality with Numbers (Assessing Skew and Kurtosis)

```
library(pastecs )
```

```
stat.desc(cbind(splicingData$SplicingFactor1,splicingData$SplicingFactor2,  
splicingData$SplicingFactor3, splicingData$SplicingEvent), basic = FALSE, norm =  
TRUE)
```

OUTPUT:

	Splicing Factor 1	Splicing Factor 2	Splicing Factor3	Splicing Event
median	5.06175631	7.66125494	4.02873177	0.7101694
mean	5.09040591	7.83867980	4.09637209	0.3985323
SE.mean	0.09128159	0.14504799	0.07599032	0.3448126
CI.mean.0.95	0.18112247	0.28780668	0.15078128	0.6841831
var	0.83323283	2.10389197	0.57745283	11.8895755
std.dev	0.91281588	1.45047991	0.75990317	3.4481264
coef.var	0.17932084	0.18504135	0.18550638	8.6520616
skewness	0.05959426	0.62923587	0.32094487	-0.1816799
skew.2SE	0.12344501	1.30341464	0.66481309	-0.3763363
kurtosis	-0.21754801	0.58418069	-0.58724752	-0.4805530
kurt.2SE	-0.22740315	0.61064464	-0.61385040	-0.5023225
normtest.W	0.99387790	0.97288592	0.97604302	0.9855238
normtest.p	0.93492888	0.03691021	0.06512656	0.3462061

SUMMARY:

Skewness:

- For Splicing Factor 1, Splicing Factor 3, and Splicing event, the p-values (0.93492888, 0.06512656, 0.3462061, respectively) suggest no significant departure from normality.
- Conversely, for Splicing Factor 2, the p-value (0.03691021) indicates a significant departure from normality and positively skewed.

Kurtosis:

- All variables Splicing Factor 1, Splicing Factor 3, and Splicing event have negative kurtosis, indicating that their tails are lighter (have fewer extreme values) compared to the normal distribution. The values are relatively close to zero, suggesting that the tails are not heavily skewed.
- Conversely, for Splicing Factor 2, Kurtosis of 0.58418069 suggest slightly heavier tail. Both measures of kurtosis are positive, indicating that the distribution has heavier tails compared to a normal distribution. The positive values suggest that there may be more extreme values in the tails.

Assessing Normality using Shapiro-Wilk normality

```
> shapiro.test(splicingData$SplicingFactor1)  
W = 0.99388, p-value = 0.9349
```

Splicing Factor 1
p-value = 0.9349 [p>0.05]: non-significant or normal

```
> shapiro.test(splicingData$SplicingFactor2)  
W = 0.97289, p-value = 0.03691
```

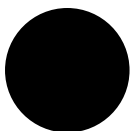
Splicing Factor 2
p-value = 0.03691 [p<0.05]: significant/ **non-normal**

```
> shapiro.test(splicingData$SplicingFactor3)  
W = 0.97604, p-value = 0.06513
```

Splicing Factor 3
p-value = 0.06513 [p>0.05]: non-significant or normal

```
> shapiro.test(splicingData$SplicingEvent)  
W = 0.98552, p-value = 0.3462
```

Splicing Event
p-value = 0.3462 [p>0.05]: non-significant or normal

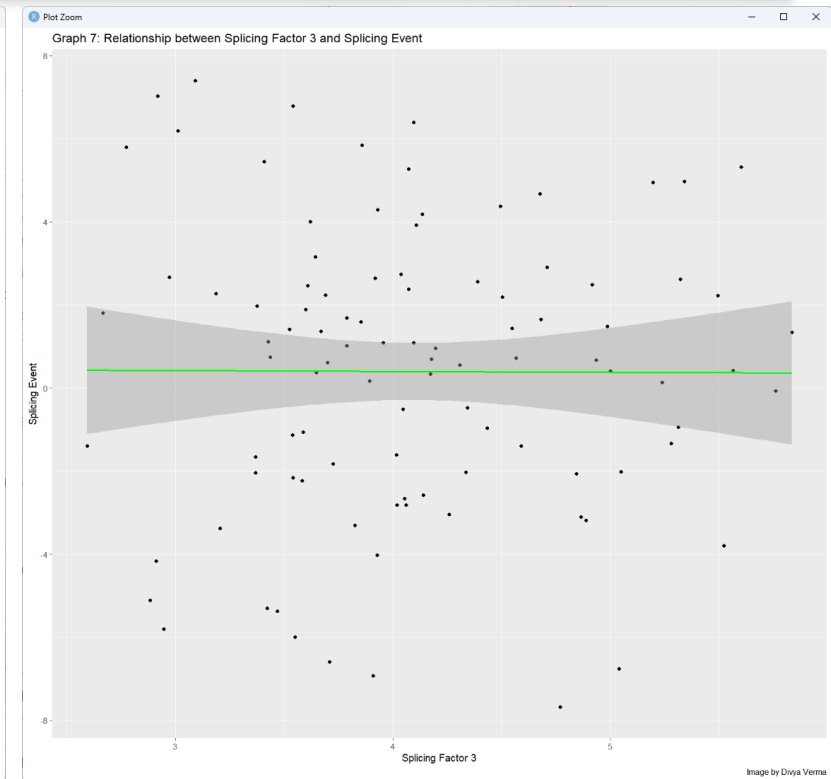
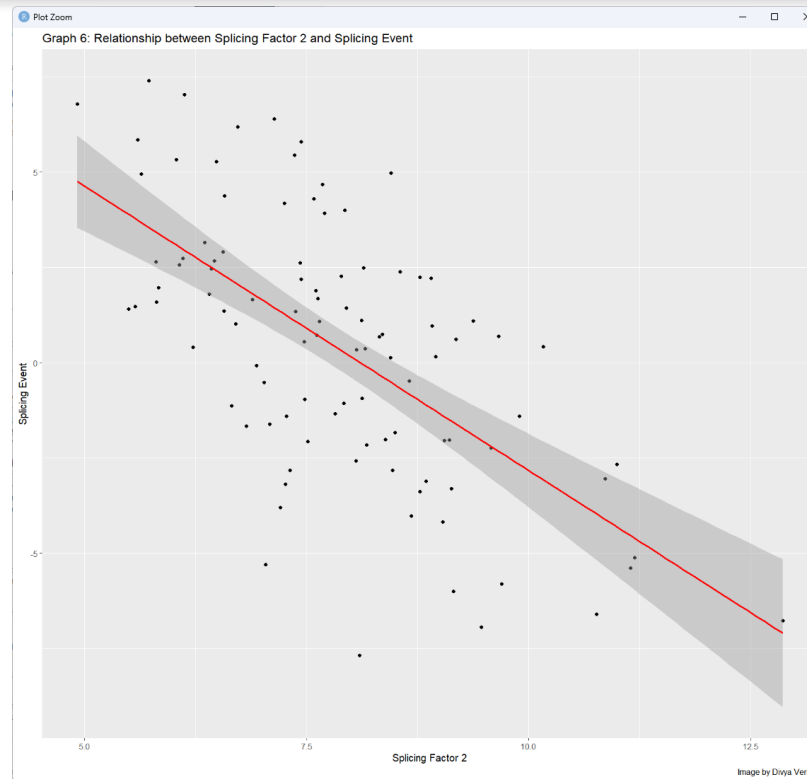
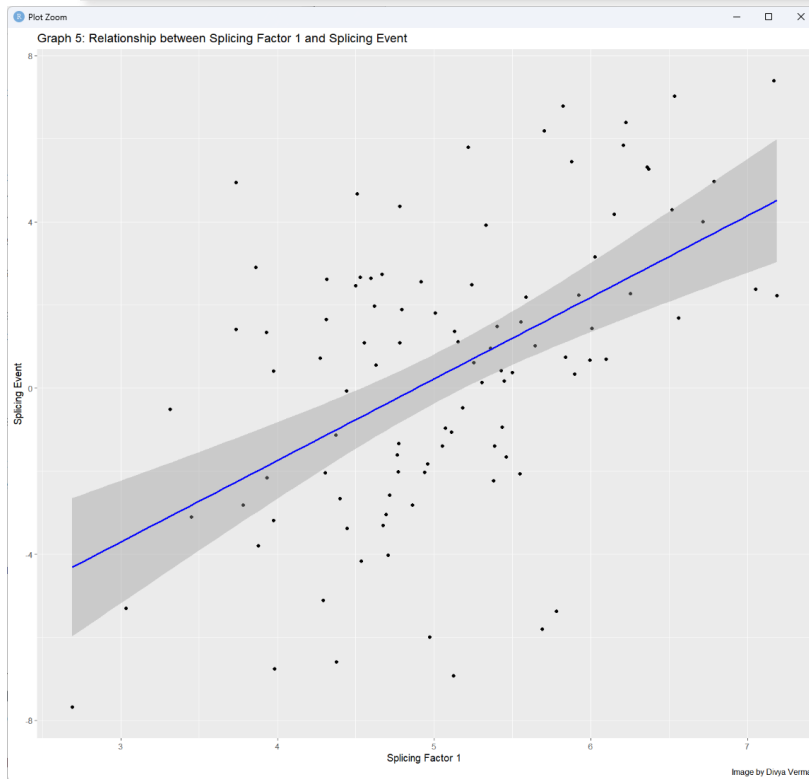


2. Data Visualization

2.2 Scatterplot to visualize the relationships between each individual splicing factor and the splicing event

RESULTS

- SplicingFactor1 is positively related to the Splicing Event.
- SplicingFactor2 is negatively related to the Splicing Event.
- SplicingFactor3 has no relation with the Splicing Event.



3. Predictive Modeling

3.1 Splitting the dataset into training and testing sets.

3.2 Building a predictive training model using linear regression.

3.3 Evaluation of model performance using appropriate metrics (e.g., R-squared, Mean Squared Error) on the testing set.

3.4 Interpret the model's coefficients to understand the relationships between the splicing factors and the splicing event.

3. Predictive Modeling

3.1 Splitting the dataset into training and testing sets

3.1.1 Create object n, which is no. of rows of splicing data

```
n <- nrow(splicingData)
```

```
OUTPUT:> n <- nrow(splicingData)> n[1] 100
```

Training set

3.1.2 Divide into 70 %

```
trainingSet <- sample(1:n, 0.7 * n)
```

3.1.3 Create train data

```
train_data <- subset(splicingData, subset = row.names(splicingData)  
%in% trainingSet)
```

SubjectID	SplicingFactor1	SplicingFactor2	SplicingFactor3	SplicingEvent
33	5.895126	8.061849	4.171631	0.34008390
9	4.313147	7.429660	5.320726	2.61551705
49	5.779965	11.150163	3.468184	-5.37924182
10	4.554338	9.378495	3.956777	1.09447048
28	5.153373	8.116941	3.426225	1.11036849
85	4.779513	7.645581	4.094516	1.09018027
56	6.516471	7.579407	3.929148	4.29621055
26	3.313307	7.022075	4.047800	-0.51510832
91	5.993504	8.321668	4.934707	0.67776805
64	3.981425	12.861560	5.040159	-6.76006468
38	4.938088	9.109921	4.335186	-2.02267439
42	4.792083	7.606704	3.598241	1.89124106
34	5.878133	7.366255	3.409178	5.44990919
43	3.734604	5.641784	5.196849	4.94706558
24	4.271109	7.615862	4.566071	0.72310091
40	4.619529	5.834160	3.374771	1.97143650
61	5.379639	9.579067	3.583106	-2.22836727

Testing set

3.1.4 Create test data

```
test_data <- subset(splicingData, subset = !(row.names(splicingData)  
%in% trainingSet))
```

SubjectID	SplicingFactor1	SplicingFactor2	SplicingFactor3	SplicingEvent
2	4.769823	8.385326	5.049930	-2.0187561
4	5.070508	7.478686	4.434555	-0.9647613
6	6.715065	7.932458	3.619002	4.0033019
8	3.734939	5.498087	3.524306	1.4088608
13	5.400771	5.573176	4.985981	1.4813085
16	6.786913	8.451730	5.340558	4.9735578
18	3.033383	7.038941	3.421547	-5.2981694
19	5.701356	6.725443	3.010982	6.1889088
22	4.782025	6.578788	4.494389	4.3792796
29	3.861863	6.557215	4.707720	2.9127990
31	5.426464	10.166826	5.564235	0.4199239
32	4.704929	8.677256	3.927744	-4.0190230
35	5.821581	4.920129	3.540489	6.7879889
39	4.694037	10.863655	4.259443	-3.0461278
41	4.305293	9.052677	3.369102	-2.0355026
44	7.168956	5.727999	3.090157	7.4030262
55	4.774229	7.820821	5.278807	-1.3297852

3. Predictive Modeling

3.2 Building a predictive training model using linear regression and interpreting its coefficients

```
TrainingModel <- lm(SplicingEvent ~ SplicingFactor1 + SplicingFactor2 +  
SplicingFactor3, data=train_data)  
summary(TrainingModel)
```

OUTPUT:

```
> summary(TrainingModel)
```

Call:

```
lm(formula = SplicingEvent ~ SplicingFactor1 + SplicingFactor2 +  
    SplicingFactor3, data = train_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.9579	-1.3056	0.0757	1.2414	4.8427

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.7162	2.4743	1.098	0.276
SplicingFactor1	1.6732	0.2862	5.845	1.71e-07 ***
SplicingFactor2	-1.4871	0.1622	-9.171	2.17e-13 ***
SplicingFactor3	0.2089	0.3411	0.613	0.542

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.083 on 66 degrees of freedom

Multiple R-squared: 0.6535, Adjusted R-squared: 0.6377

F-statistic: 41.48 on 3 and 66 DF, p-value: 3.465e-15

Splicing Factor1 has a positive coefficient estimate of 1.6732, indicating that an increase in SplicingFactor1 is associated with an increase in the Splicing Event.

Splicing Factor2 has a negative coefficient estimate of -1.4871, indicating that an increase in SplicingFactor2 is associated with a decrease in the Splicing Event.

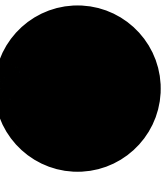
Splicing Factor3 has a coefficient estimate close to zero (0.2089) and a high p-value (0.542), suggesting that its effect on the Splicing Event **may not be statistically significant**.

The **residual standard error** represents the standard deviation of the residuals around the regression line. In this case, it is approximately 2.083.

The **Multiple R-squared** measures the proportion of variance in the Splicing Event variable explained by the predictor variables. In this case, it is 0.6535, indicating that the predictor variables explain about 65.35% of the variance in Splicing Event.

The **Adjusted R-squared** adjusts the Multiple R-squared for the number of predictor variables in the model. In this case, it is 0.6377.

The **F-statistic tests** the overall significance of the regression model. It compares the variability explained by the model to the variability not explained. A **low p-value** associated with the F-statistic indicates that at least one predictor variable is significantly related to the outcome. In this case, the p-value is very low (3.465e-15), indicating that **the regression model is statistically significant**.



3. Predictive Modeling

3.3 Evaluate the model's performance using appropriate metrics on the testing set

R-squared

```
# Evaluate the model's performance on the testing set.
```

```
predictedTest <- predict(TrainingModel, newdata =  
test_data)
```

```
#Calculate R-squared
```

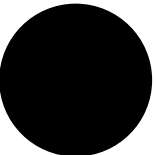
```
R_squaredTest <- cor(predictedTest,  
test_data$SplicingEvent)^2
```

```
> R_squaredTest [1] 0.6954288
```

Interpreting R-squared:

Goodness-of-fit:

- A R-squared value 0.6954288 ($0.5 \leq R^2 < 0.7$) indicates a moderate to strong level of goodness-of-fit.
- It suggests that the regression Training Model captures a significant portion of the variation in the dependent variable or the Splicing Event.
- So, it has a good prediction accuracy.
- Ideally this value should be $R \geq 0.7$ for a strong explanatory power



3. Predictive Modeling

3.3 Evaluate the model's performance using appropriate metrics on the testing set

Mean Squared Error

Calculate Mean Squared Error (MSE)

```
MSE_Test <- mean((predictedTest -  
test_data$SplicingEvent)^2)MSE_Test
```

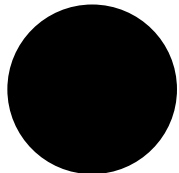
OUTPUT

```
> MSETest
```

```
[1] 3.105155
```

Interpreting Mean Squared Error:

- In general, a lower MSE indicates better predictive performance of the test model, as it represents smaller prediction errors.
- Ideally the MSE should be close to zero.
- A Mean Squared Error (MSE) of 3.105155 means that, on average, the squared difference between the predicted values and the actual values in the dataset is approximately 3.105155.
- A higher MSE suggests that the model's predictions are less accurate and have larger errors compared to the actual values.



4. Conclusions and Recommendations

4.1 Summary of key findings from EDA and modeling.

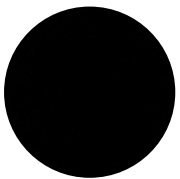
4.2 Identification of splicing factors most strongly associated with the splicing event.

4.3 Recommendations for further analysis or experiments based on results.

4. Conclusions and Recommendations

4.1 Summary of key findings from EDA and modeling.

- No missing values were found in the dataset, but two Outliers were detected- one in SplicingFactor1 and one in SplicingFactor2.
- Data is mostly symmetrical for SplicingFactor1, SplicingFactor3, and SplicingEvent where as SplicingFactor2 has significant departure from normality and is positively skewed.
- Predictive Modeling using Linear regression:
 - The regression analysis on the model showed that the model is overall statistically significant.
 - Model performance metrics on the test model:
 - **R-squared** of 0.6535 indicates a moderate to strong level of goodness-of-fit
 - Mean Squared Error (MSE) of 3.105155 suggests that, on average, the model's predictions have a high error of approximately 3.105155 units.

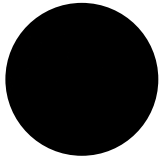


4. Conclusions and Recommendations

4.2 Identification of splicing factors most strongly associated with the splicing event.

Correlation Analysis:

- SplicingFactor 1 is positively related to the Splicing Event.
- SplicingFactor 2 is negatively related to the Splicing Event.
- SplicingFactor 3 has no relation with the Splicing Event.
- SplicingFactor 2 is most strongly associated with the splicing event and negatively correlated with SplicingEvent ($r = -0.63$), and the significance value, $p\text{-value} = 0$, indicating a significant negative relationship.
- SplicingFactor 1 is positively correlated with SplicingEvent ($r = 0.52$), and $p\text{-value} = 0$ indicating a significant positive relationship.

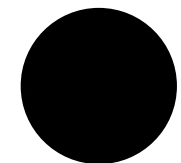


4. Conclusions and Recommendations

4.3 Recommendations for further analysis or experiments based on results.

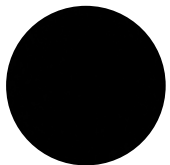
The presence of outliers in Splicing Factor 2 is contributing to significant skewness and kurtosis in the data. Given that Splicing Factor 2 is highly negatively correlated with the splicing event, these outliers may have a big impact on the results of statistical modeling.

To address this issue and obtain more accurate results, it is recommended to remove the outliers from the dataset and rerun the statistical modeling. By eliminating the outliers, the skewness and kurtosis of the data distribution can be reduced, potentially leading to a better fit of the model and more reliable estimates of the relationship between Splicing Factor 2 and the splicing event.



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Thank You

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